Email

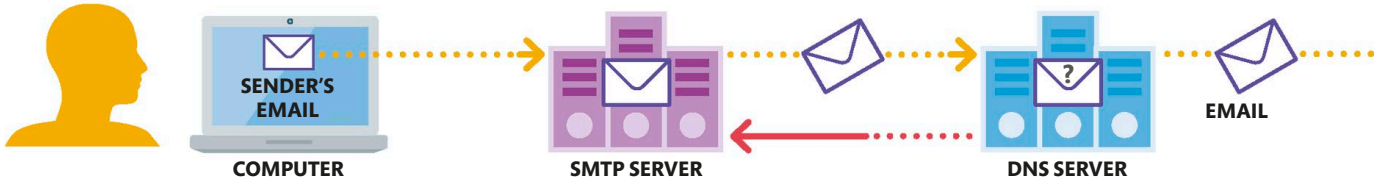
Electronic mail (email) is a method for exchanging messages using computers and other devices. Connecting to an email server allows a user to send and receive messages as well as other files in the form of attachments.

WHO SENT THE FIRST EMAIL?

Ray Tomlinson is credited with sending the first email, in 1971. While working for ARPANET, he developed a way for messages to be sent between computers.

How emails are sent

Emails are exchanged according to a set of rules, the Simple Mail Transfer Protocol (SMTP), which allows for communication across different devices and servers. When a user sends an email, the message is uploaded to an SMTP server, which communicates with the domain name system (DNS) to check the recipient's server address before delivery. An Internet domain is a group of addresses under the control of an individual or organization.



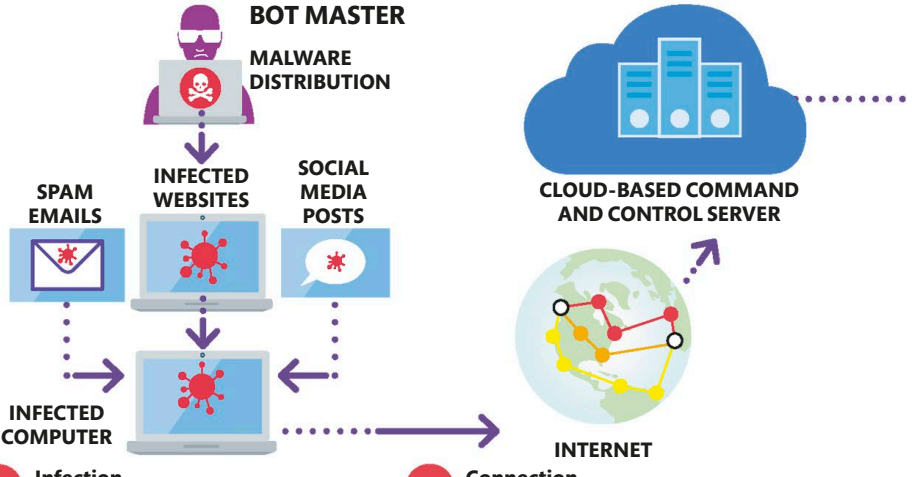
1 Email sent
The sender composes a message using a mail client: an application for composing, sending, and reading emails. He or she also enters the recipient's email address. Once the user presses send, the delivery process begins.

2 SMTP server
The message is sent to an SMTP server—the online equivalent of a post office. On this server, a mail transfer agent (MTA) checks the recipient's address and then looks up its domain.

3 DNS server
The MTA must communicate with a DNS, which translates domain names into IP addresses. The recipient's domain is checked to find his or her mail server. If it cannot be found, an error message is returned.

Spam and malware

Sending emails is cheap, so email is often used to send content to many users at once. Some unsolicited emails (called spam) are merely annoying, while others spread deliberately destructive software (malware). Once malware has been downloaded, it may disable, hijack, or alter computer functions, monitor activity, demand payments, encrypt or delete data, or spread to other computers. Email filters scan incoming emails for content indicating spam and malware.



How a botnet works
Hackers who wants to anonymously carry out malicious activities online may breach the security of connected devices to create a network of devices they control: a botnet.

1 Infection
Hackers use malware that contains bots: applications that perform automated tasks. The malware is distributed, and if it is downloaded, it can infect the user's computer.

2 Connection
The bots discreetly instruct the infected computer to connect to a command and control (C&C) server. The hackers use this server to monitor and control the botnet.